EXPERIMENT NO:07

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# Generate dataset
X, y = make_classification(
    n_samples=200,
    n_features=2,
    n_redundant=0,
    n_informative=2,
    random_state=42
)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train model
model = LogisticRegression()
model.fit(X_train, y_train)
```

```
# Prediction
y_pred = model.predict(X_test)

# Display results
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", round(accuracy * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Graph 1: Classification graph
plt.figure(figsize=(7, 5))
plt.scatter(X[:, 0], X[:, 1], c=y, cmap='coolwarm',
            edgecolors='black')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Logistic Regression Classification")
plt.colorbar(label="Class")
plt.grid(True)
plt.show()

# Graph 2: Actual vs Predicted
plt.figure(figsize=(8, 4))
plt.plot(y_test, 'o-', label="Actual")
plt.plot(y_pred, 'x--', label="Predicted")
plt.xlabel("Test Sample")
```

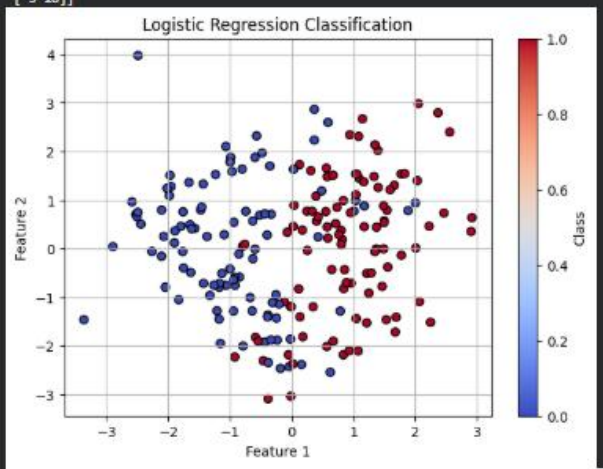
```
plt.ylabel("Class")
plt.title("Actual vs Predicted Classes")
plt.legend()
plt.grid(True)
plt.show()

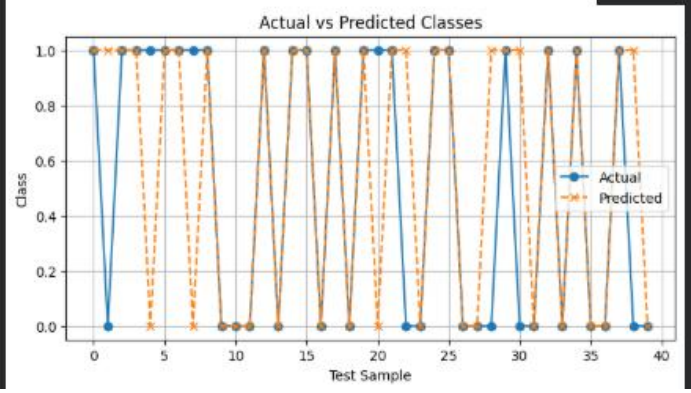
# Graph 3: Accuracy
plt.figure(figsize=(5, 4))
plt.bar(["Logistic Regression"], [accuracy * 100])
plt.ylim(0, 105)
plt.ylabel("Accuracy (%)")
plt.title("Logistic Regression Accuracy")
plt.text(0, accuracy * 100 + 2,
        f"{accuracy * 100:.2f}%", ha='center')
plt.show()
```

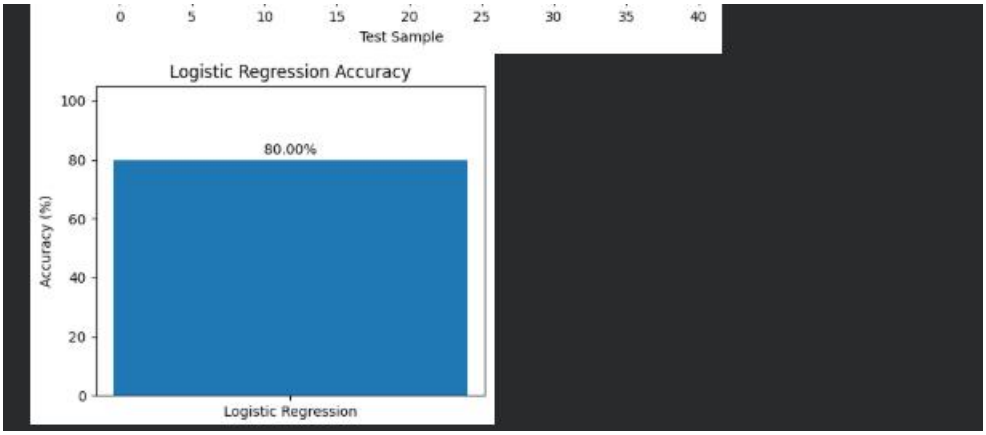
OUTPUT:

```
Accuracy: 80.0 %

Confusion Matrix:
[[14  5]
 [ 3 18]]
```







```
*** Accuracy: 96.67 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]
```

